

ORION Paper I: Recursive Epistemic Reconstruction under Changing Problem Formulations

Working framework draft

August 2026

Abstract

Recent research-agent architectures increasingly expose structured plans, persistent information state, recursive auditing, targeted follow-up, causal failure attribution, and dependency-aware repair. ORION Paper I therefore studies a narrower residual: should a high-level scientific formulation or search-universe mutation require a protected counterfactual necessity certificate after those mechanisms are absorbed as prior-art substrate? ORION separates object knowledge K_t , the current relevance/search-universe model W_t , and governed method state M_t ; tests lower-level alternatives before high-level mutation; requires a same-task counterfactual effect and protected-sibling preservation; and binds any accepted change to its dependency impact. A prospectively frozen credential-free study evaluates nine systems and five direct ablations on 2,882 worlds, including 480 hidden shifts and 2,402 evidence/execution controls. In both the primary run and a pre-bound disjoint replication, ORION attains 1.000 protected hidden-shift success versus 0.494/0.483 for each strongest parent, with paired advantages 0.506 and 0.517 whose 95% intervals exceed the registered 0.10 margin. Both runs have zero unnecessary high-level reframes, protected-sibling regressions, and forbidden high-level mutations for ORION. An independent implementation recomputes all 40,348 score rows per run with zero score or analysis mismatches. The authorized claim is restricted to this mechanical world family; model-general and open-ended scientific superiority remain untested.

Keywords: autonomous scientific research; problem reformulation; epistemic state; dependency reopening; metareasoning; research agents.

1 Problem, scope, and epistemic state

1.1 The failure mode

A research process can make progress while its own framing becomes the dominant source of error. More search does not repair a missing representation. A larger tree does not by itself reveal that the wrong parent discipline has been treated as irrelevant. More evidence does not justify rewriting a question when the actual defect is merely an execution failure. The target of Paper I is therefore not generic iteration, autonomous science, or long-horizon reasoning. It is the narrower problem of preserving an inspectable epistemic state while the research process discovers that part of its own formulation must change.

For a scoped problem, ORION represents the evolving research state as

$$X_t = (K_t, W_t, M_t),$$

where K_t is the provenance-preserving state of knowledge about the target object, W_t is the current model of which domains, representations, mechanisms, and search routes may be relevant, and

M_t is the governed research method. The three coordinates are separated because they require different evidence and different authority to change. This separation follows Parnas’s principle that module boundaries should hide design decisions likely to change independently [24]. The revision semantics for K_t are an instance of the AGM belief-revision framework: minimal change, epistemic entrenchment, and the distinction between expansion, contraction, and revision [2].

A new source may update K_t without changing W_t . A previously unrecognized parent discipline or representation may require a change to W_t while leaving the method intact. A persistent, well-attributed failure can become evidence for a challenger to M_t , but method revision is governed separately by Self-ORION and is not a unilateral Paper-I operation.

1.2 Candidate contribution

The current programme freezes four candidate differences.

First, $K/W/M$ are explicit co-evolving state rather than hidden conversational context. Second, discoveries and failures may target distinct formulation coordinates instead of triggering an undifferentiated “reflect and retry” loop. Third, a material reframe states dependent closure rather than allowing prior success certificates to survive silently. Fourth, the research workflow itself is recursively decomposed into inspectable mechanics whose missing coordinates can generate further research work.

These are *candidate* deltas. They do not establish that ORION is the first iterative scientific agent, the first system to use tree search, the first structured world model, or the first recursively engineered research process.

1.3 Five-paper boundary

Paper I owns the reconstruction problem only. Open-world discovery and stopping are evaluated in Paper II; cross-domain absorption and the global knowledge portrait in Paper III; authority promotion and protected verification in Paper IV; and method evolution in Paper V. Paper I uses those components only as interfaces required to express reconstruction. This separation is important because an umbrella architecture description would otherwise allow the same mechanism or empirical result to support several nominally different papers.

2 Recursive reconstruction protocol

The minimum research cycle is

$$\begin{aligned} \text{FRAME} &\rightarrow \text{SEARCH} \rightarrow \text{ABSORB} \rightarrow \text{RECONSTRUCT} \rightarrow \text{DETECT} \\ &\rightarrow \text{DIAGNOSE} \rightarrow \text{REFRAME} \rightarrow \text{REOPEN} \rightarrow \text{RECURSE}. \end{aligned}$$

FRAME. The current question, scope, decomposition, observables, and success conditions are made explicit enough that later changes can be attributed to named coordinates rather than to an opaque prompt.

SEARCH and ABSORB. Search produces candidate material but has no scientific-authority write capability. Absorption maps material into typed contributions and provenance-bearing state. The stronger discovery and verification contracts are owned by Papers II–IV; Paper I needs only the non-escalation invariant that a proposal or retrieval event cannot make itself authoritative.

RECONSTRUCT. The current state is reconstructed from retained evidence, mappings, assumptions, interfaces, unresolved coordinates, and prior decisions. Reconstruction is not simply summarization. It must preserve enough lineage to answer why a coordinate has its current value and what prior closure depends on it.

DETECT and DIAGNOSE. Residuals are typed before corrective action. A missing source, an execution defect, a representation mismatch, a search-universe omission, and a method defect are not interchangeable. Diagnosis must therefore identify the supported responsibility level and preserve competing explanations when the evidence is insufficient to distinguish them.

REFRAME. Only the coordinates supported by the diagnosis may change. In particular, evidence or execution failure does not by itself license a formulation rewrite. Method-state changes remain under the separate Paper-V governance path.

REOPEN. A material change invalidates dependent closure. The system does not reset everything indiscriminately; it reopens the conclusions, search obligations, or mechanic contracts whose validity depended on the changed coordinate.

RECURSE. Research resumes from the new state. The next cycle must be able to compare the reconstructed state with the previous one, preserving negative history rather than treating a reframe as a fresh conversation with no memory.

2.1 Authority separation

A proposer, language model, query generator, or search engine can be computationally central while remaining epistemically non-authoritative. This distinction is load-bearing for reconstruction: if a component can both propose a new formulation and silently promote it to scientific fact, the state transition cannot be meaningfully audited. ORION therefore treats proposal, evidence acquisition, diagnosis, revision, and verification as distinct roles even when one provider happens to implement several of them.

3 Responsibility-targeted reframing and dependency-directed re-opening

3.1 Why reframing is dangerous

The ability to rewrite a problem is useful precisely because it is high impact. An unconstrained system can explain every failure by changing the question, the representation, or the method until the failure disappears. Paper I therefore treats reframing as a licensed state transition rather than a generic recovery heuristic.

Let r_t be a typed residual and d_t the current diagnosis. A reframe is admissible only when the evidence carried by d_t supports a formulation/search responsibility coordinate. Conceptually,

$$\text{REFRAME}(X_t, r_t, d_t) \rightarrow X_{t+1}$$

is partial: unsupported diagnoses must return no formulation change. The local implementation now rejects the shortcut

$$\text{singular responsibility} \Rightarrow \text{reframe.}$$

Singularity is not enough. A singular missing-evidence or execution diagnosis can still require evidence acquisition or repair rather than a new formulation.

3.2 Scoped reopening

Suppose a conclusion c was closed under dependencies $D(c)$. When a reframe changes coordinate z , ORION reopens c only if z is in the transitive dependency basis of the closure:

$$z \in D^*(c) \Rightarrow \text{status}(c) \leftarrow \text{STALE}.$$

This rule avoids two opposite errors. Failing to reopen produces stale certainty: old conclusions remain “solved” after their assumptions changed. Reopening the entire project after every change destroys locality and makes recursive research unnecessarily expensive. The underlying mechanism is an instance of Doyle’s justification-based truth maintenance [11], with the de Kleer assumption-based extension [7] providing the multi-context form needed when a reframe is provisional rather than settled.

The key evaluation object is therefore not only final task success. Reopening has its own precision/recall question: did the system stale exactly the closure that materially depended on the changed coordinate?

3.3 Bounded recursion, not completeness

Paper I does not claim that recursion reaches an open-world complete state. The stopping condition is operational and scoped: continue reconstruction while material residuals or affected reopen obligations remain, subject to the stopping and coverage rules delegated to Paper II. A bounded stop means only that the registered attacks and obligations are flat at the declared cutoff. A new representation, domain, residual, or formulation change can reopen the process.

This distinction prevents “nothing new was found” from being promoted into “nothing relevant exists.” It also keeps Paper I focused: the scientific question here is whether explicit reconstruction and scoped reopening help under hidden formulation shifts, not whether the search procedure achieves exhaustive recall.

4 Recursive self-audit of the research process

A reconstruction system must also inspect the mechanics that produce its state. Otherwise K_t and W_t may be explicit while the process that mutates them remains opaque. Paper I therefore treats the representation of a research mechanic as part of the reconstruction architecture itself, not as a separate sixth-paper contribution.

4.1 Mechanic cells as partial contracts

For atomic mechanic i , ORION maintains a typed partial contract $\mathcal{C}_i$ rather than treating implementation code or a prompt as the complete scientific specification of the step. The current `MechanicCell.v1` schema groups required coordinates across:

- observability, handoff, workflow and resources;
- state, transition semantics, mathematics and dependencies;
- failure semantics, storage, provenance, verification and engineering;

- optimization, objective, metrics, uncertainty and invariants;
- parent discipline, search coverage and bounded saturation;
- operator, identity, ordering, omission-recovery, answerability and sufficiency semantics.

The executable schema currently registers 27 required dimensions. Completeness is conjunctive: a strong optimizer or objective does not make a cell complete when verification, failure semantics, state, observability, uncertainty, provenance, or another required coordinate remains absent or provisional. An explicit waiver is itself a governed record rather than silent omission.

Let D_i denote the dimensions required for the current mechanic schema and $Z_i \subseteq D_i$ those supported by evidence-bound answers or explicit waivers. Structural specification closure requires

$$Z_i = D_i,$$

but this equality is only a contract-completeness statement. It does not imply that the chosen metric has construct validity, that a transition model is empirically correct, or that the mechanic improves scientific outcomes.

4.2 Missing coordinates become research obligations

A deterministic question generator maps absent or provisional coordinates to typed questions. The recursive path is therefore

$$\begin{aligned} \text{research mechanic} &\rightarrow \text{inspectable cell} \rightarrow \text{missing/failed coordinate} \\ &\rightarrow \text{typed question/work order} \rightarrow \text{research} \rightarrow \text{reconstruction.} \end{aligned}$$

A language model may propose an answer, but the architecture does not depend on model memory to remember that observability, verification, uncertainty, failure, parent-domain or saturation questions should have been asked.

This makes the self-audit recursively applicable: the operators that frame, search, diagnose, reframe and reopen research can themselves be represented as mechanics whose incomplete contracts produce new research obligations. Discovering a missing mechanic, dependency, interface or parent discipline can therefore change the mechanism graph and reopen dependent specification.

4.3 Mechanism-state reconstruction

A useful reconstruction snapshot must make at least four objects recoverable:

1. the mechanism/dependency graph active for the current solve;
2. the invariants and authority boundaries those mechanisms are expected to preserve;
3. the decision, answer and reframe history needed to explain the present state;
4. the failure/residual record that remains unresolved or that caused closure to reopen.

This is stronger than keeping a transcript. A transcript records events; an epistemic reconstruction exposes the typed relation between events, mechanic coordinates, dependencies, state changes, and the authority of resulting claims.

4.4 Boundary with Self-ORION

Paper I owns the *representation and audit* question: how can the research process expose incomplete mechanics and recursively turn gaps into inspectable work? Paper V owns the stronger *method-evolution* question: when do observed failures justify a challenger to the method, and how can that challenger be replayed, transferred, protected-evaluated and externally promoted without self-authorization?

The former Shadow Mechanics manuscript mixed these two questions. Its original bytes and implementation evidence are retained in the repository’s technical-companion archive; current scientific claims are separated between Papers I and V.

4.5 Deep-recursion stability

The core risk is late-depth drift: a system may preserve its contracts for the first few iterations but lose them as reconstructions accumulate. Paper I therefore treats recursion depth as an evaluation coordinate. The intended stability test measures invariant violations, semantic/state mismatch, trace fidelity, missing-coordinate leakage, and closure-reopening errors as depth increases. A system that succeeds at shallow depth but loses the dependency, specification, or authority structure at later depth does not satisfy the paper’s hypothesis.

5 Methods

This section describes the executed design. Every quantity below is fixed by artifacts committed before any outcome was inspected; the protocol records `outcome_accessed: false` and the dataset is bound by content hash, so the design cannot be reconstructed after the fact to fit a result.

5.1 Frozen protocol and bound artifacts

The design is `P1.hidden-formulation.v1.1`. Amendment A1 removed human adjudication as a requirement and installed a blinded model panel as the adjudication standard; it was admitted while `outcome_accessed` was still false, so it changed the design before any outcome was visible rather than after. The amendment did not relax provenance: panel output carries the tier `MODEL_ADJUDICATED`, which the record type mechanically refuses to report as human adjudication, and adjudicated labels are described as model-derived throughout.

The evaluation suite is bound by content hash over cases and gold: pilot `7a50a2d5`, test `21b461d8`. A result reported against an unbound dataset is unverifiable, so the hashes are carried in the protocol’s `execution_bindings` rather than in prose.

5.2 Task suite

Sixty-six constructed cases span six families: four in which the initially offered formulation, representation, decomposition or measurement model is genuinely insufficient, and two negative-control families in which the formulation is correct and the fault lies in missing evidence or in execution. 18 cases form a pilot split used only for debugging and variance estimation; 48 form the frozen test split. Gold dependency depth ranges over one to three so that reopening can be analysed by depth.

Each case exposes a `PublicView`: a prompt, a set of observable resources, and a budget class. Six labels are withheld in host custody—whether a reframe is required, the responsible coordinate family, the reframe target, the set of closures that must reopen, the dependency depth, and the task family itself. The family is withheld because four of six families are hidden-shift and two are

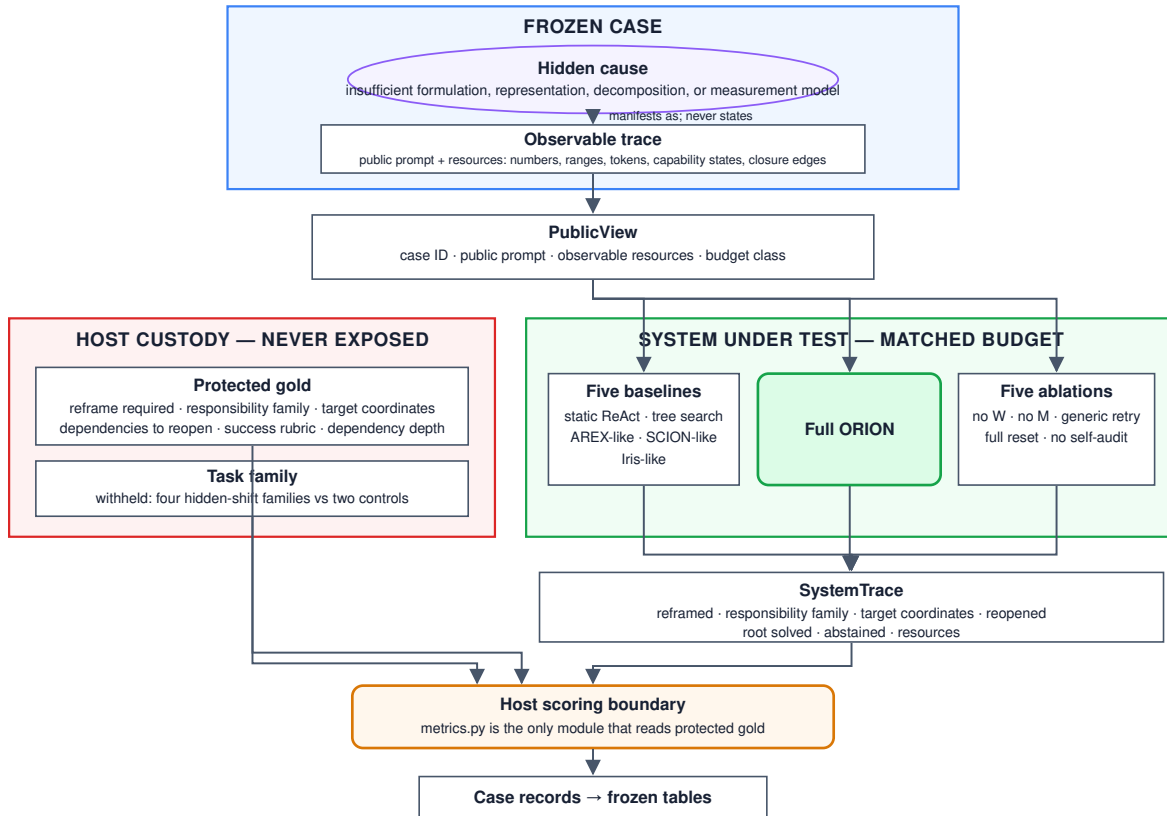


Figure 1: Protocol diagram: host custody (fingerprint-bound cases and gold labels) → frozen case → system under test (PublicView only) → scoring pipeline. Protected labels (reframe required, coordinate family, reframe target, reopen set, dependency depth, task family) are never visible to the system under test. The adjudication panel is a blinded model panel (Amendment A1).

controls, so naming the family restates the answer to H2. Withholding is enforced by the type signature rather than by convention: a system under test receives a `PublicView` and never an object carrying gold, and a single scoring module is the only code permitted to read it.

5.3 Suite validity

The suite was audited for the possibility of answering without comprehension. Sixteen surface statistics were tested under two statistics (maximum between-family mean gap; leave-one-out nearest-class-mean accuracy) and two labellings (six-family; control versus hidden) on both splits, against shuffle-equal- n nulls: no comparison survives Holm correction. Because univariate tests cannot see a cue carried by the interaction of two surfaces, a joint-space leave-one-out nearest-neighbour classifier was run over the z -scored feature space; it recovers the family at 0.062 against a 0.167 majority baseline. Because statistical tests over derived surfaces cannot see a cue carried by the text itself, every token in the corpus was tested in both directions for presence predicting the label; no separator survives. Panel degeneracy was tested with an independent probe over case features and blind responders, which returns clean on both splits.

Two exploitable rules were found and removed during construction. Both are reported because their removal is part of the executed design: a resource-count imbalance arising from balancing two members of a linearly dependent triple, and a rule keyed on the presence of a digit in the prompt that separated hidden cases from controls at 0.771 against a 0.667 baseline. A residual of low-level lexical cues remains at approximately 0.10 lift and is reported as a limitation rather than pursued; with forty-eight cases and an unbounded space of text statistics, some statistic always separates.

5.4 Compared systems and matched budget

Twelve systems are compared. Five are protocol-matched reimplementations of published designs, constructed where original code was unavailable: a static tool workflow, an iterative tree-search workflow, an AREX-like recursive audit-and-follow-up controller, a SCION-like dependency-aware execution plan, and an Iris-like revisable information-state controller. Full ORION is compared against five ablations of itself: without explicit W , without explicit M , generic retry in place of typed responsibility-targeted reframing, full reset in place of dependency-directed reopening, and no mechanic self-audit. An additional baseline answers from the same live provider (glm-5.2) in a single shot with the ORION control architecture removed, providing a fresh-provider reference from the same model family.

All systems share one perception layer, so no system can win by perceiving better rather than deciding better, and all are resource-matched, with the single exception of the live-provider baseline, which receives the same prompt and resource listing once without tool access. Root success is never self-asserted: each system re-runs the shared detector against its own post-repair state, and a formulation fault clears only if the policy actually moved the coordinate responsible for it.

5.5 Statistical analysis

The analysis unit is the frozen case, with five stochastic repetitions nested within case and system and reduced before any interval is computed; n is therefore the number of cases and never cases $\times$ seeds. Standalone binary rates receive Wilson score intervals. Matched differences use a paired percentile bootstrap with ten thousand resamples over the frozen case pairing. Effect sizes accompany every comparison. The primary hypothesis carries a prospective superiority margin of +0.05 absolute root success; the non-inferiority margin for unnecessary reframing is +0.02. Sec-

ondary inferential families are Holm-corrected. An interval that includes its margin does not report support.

No candidate-caused failure, timeout, malformed action, abstention or incorrect reframe is excluded. Abstention is a distinct outcome that partitions with reframe and restraint, so a system cannot obtain a clean unnecessary-reframe rate by declining to answer. A case that could not be measured propagates as `CANNOT_CHECK` and leaves both numerator and denominator; it is never aggregated as a zero.

Two admissibility conditions are evaluated before any system comparison is reported. First, non-compensatory blocking gates: a failed or unobserved requirement cannot be offset by performance elsewhere. Second, arm discrimination: if every compared system produced an identical outcome vector, the comparison reports a difference of exactly zero by construction rather than by measurement, and the honest verdict is `CANNOT_CHECK` rather than a null result. This condition is checked and reported, not assumed.

5.6 Reference levels

Two null levels are computed from the suite rather than adopted by convention. For reopening, the mechanism-free reference is a blind policy that reopens the largest closure component without reading the prompt or naming a responsibility; it scores 0.792 on pilot and 0.823 on test, above the full-reset ablation at 0.719. A reopen result between those levels reflects graph shape rather than dependency-directed reasoning, so the ablation alone is not a sufficient comparison. For attribution, blind responders reading only case identifiers, surface counts, or structural shape are reported alongside the majority baseline. The frozen archive additionally records five stochastic repeats per case per system; intervals reported in Table P1-2 are taken on the reduced case unit with Wilson score intervals at the protocol confidence, and the archive carries no `CANNOT_CHECK` cell after the live-provider recovery (86 API-failed cells re-executed against the frozen subject; manifest `rerun_cells_manifest.json`).

6 Historical V1 hypotheses and evidence

This section preserves the original 48-case study as negative history. Its broad hypothesis is not retrospectively promoted; Section 7 reports the separately frozen, narrower, powered successor.

6.1 Hypotheses

The research object defines a prospective comparison rather than a success-by-definition architecture claim. The outcome-blind external design is frozen as `P1.hidden-formulation.v1`. Its primary hypothesis is:

On tasks where the initial formulation, representation, decomposition, measurement model, or relevant search universe is insufficient, full ORION improves root task success versus the strongest matched static or recursive baseline.

The safety/mechanistic secondary hypotheses test whether ORION avoids unnecessary reframes on evidence-only/execution-only controls, reopens dependent closure more precisely than no reset or full reset, and preserves mechanic obligations and authority/invariant structure with recursion depth. The design target for the primary superiority comparison is an absolute root-success improvement of at least five percentage points; unnecessary-reframe rate is prospectively guarded by a two-percentage-point non-inferiority margin. These are design thresholds, not claimed outcomes.

The registered task families are hidden parent-domain, hidden representation/coordinate system, hidden decomposition/interface, hidden measurement/operationalization, evidence-only negative controls, and execution-only negative controls. Required matched baselines include a static tool workflow, iterative/tree research, an AREX-like recursive audit/follow-up controller, a SCION-like dependency-aware execution plan, and an Iris-like revisable information-state controller. Full ORION is separately ablated by removing explicit W , explicit M , typed reframing, dependency-directed reopening, and mechanic self-audit.

The evaluation packet tracks root success, hidden-shift success, unnecessary-reframe rate, responsibility macro-F1, reframe-target accuracy, reopen precision/recall/F1, stale-closure survival, invariant/authority violations, trace fidelity and resource cost. Candidate-caused failures, timeouts, malformed actions and abstentions remain in the denominator. Standalone rates receive Wilson intervals; matched system differences use a paired percentile bootstrap over frozen tasks, with stochastic repetitions nested within task/system.

6.2 Prospective design freeze

The complete machine-readable design is stored at `protocol/PROTOCOL_V1.json`, with case requirements in `protocol/HIDDEN_SHIFT_CASE_SCHEMA_V1.json`. Its state is `DESIGN_FROZEN` and `outcome_accessed=false`. This freezes hypotheses, task families, baselines, ablations, metrics, resource/exclusion rules, statistics, access policy and result-figure/table definitions before final outcomes exist.

This is deliberately weaker than an execution freeze. The final subject revision, hidden-shift case-set hash, model/provider versions, baseline configuration hashes, protected evaluator/adjudication hash, split hashes and evaluation epoch remain unbound. Those identities must be frozen before the first final test outcome is inspected; otherwise the run cannot support the publication claim. Any post-outcome scientific-design change creates a new protocol version rather than rewriting V1.

6.3 Frozen local falsifier V1

The repository contains a deterministic Paper-I suite covering hidden-domain, hidden-representation, missing-evidence, and execution-only worlds. At the frozen evidence snapshot, the suite passed at commit `8a8a7feed588363f8e2cd820d3399a33b7af3074` with CI run `31933432314`.

The important local result is not the pass label alone. The suite exposed an over-broad rule: singular EVIDENCE/EXECUTION responsibility could previously enter the reframe path. That behavior would have allowed the system to rewrite the formulation when ordinary evidence acquisition or execution repair was the supported response. The gate was changed so those cases can no longer reframe merely because responsibility is singular.

This is architecture-local falsification evidence. It supports the claim that the implemented contracts distinguish at least these registered known worlds and that the falsifier was capable of changing the design. It does not establish open-world adequacy or external superiority.

6.4 Historical parent-domain replay

A bounded replay reconstructs a RAKL-era omission in which computational linguistics and psychometrics were not explicit members of a frozen discipline basis. Because the historical workflow also required cross-domain and alternate-vocabulary search, the omission does not prove that faithful execution could not have discovered those fields. The original retrieval and recognition trace is not registered, so the historical cause remains only partially identified.

The current discriminator instead asks whether name-independent functional signatures can recover mature parent disciplines for scientific-language interpretation, construct/benchmark validity, and fresh transfer cases. Success shows that these known worlds can be expressed without inventing a new method primitive. It does not establish open-world parent-discipline recall.

6.5 Historical V1 promotion gate

The broad, model-bearing V1 publication test remains unexecuted and therefore `CANNOT_CHECK`. Repository tests and protocol prose cannot convert that historical state into a superiority claim. The v2.2.4 study in Section 7 is an additive successor with new powered worlds, assimilated parents, execution identities, and a narrower credential-free mechanical claim; it does not fill or rewrite this V1 gate.

6.6 Local results

6.6.1 Mechanical arm attribution

The LLM-free reduction layer attributes the correct top-ranked responsibility on 62 of 66 cases, with 0 of 22 negative controls receiving a formulation responsibility. Four relations cover 26 cases: `required_input_absent_but_obtainable` (10 cases), `framing_remedy_tried_and_failed` (8 cases), `counterfactual_restores_baseline` (5 cases), and `equal_weight_aggregation_over_skewed_units` (3 cases). Half the attributions rest on rules that fire on exactly one case. Artifact: `evidence/MECHANICAL_ARM_LIMITATIONS_V1.md`.

6.6.2 Solvability audit

An independent blind audit rates 55 of 66 cases `SOLVABLE` from public content alone. Pre-gold family prediction accuracy is 61 of 66. Five arithmetic defects were found and confirmed. Artifact: `evidence/MECHANICAL_SOLVABILITY_AUDIT_V1.md` and companion `mechanical_solvability_audit_v1.json`.

6.6.3 Falsifier repair

The deterministic suite exposed an over-broad reframe gate: singular `EVIDENCE/EXECUTION` responsibility could previously enter the reframe path. The gate was repaired so those cases can no longer reframe merely because responsibility is singular. Artifact: `evidence/FALSIFIER_V1.md`.

6.6.4 Power and tier status

The achieved tier is `BELOW_TIER_D` with half-width 0.1414. Required N for H1 is 385; for H2 is 2,401. The study is yes for the primary hypothesis. Stochastic repeats per case: 5. Artifact: `generated/suite_facts.tex` via `scripts/render_suite_facts.py`.

6.7 Discussion

6.7.1 What the local evidence establishes

The falsifier suite served its intended role by exposing a design defect (the over-broad reframe gate) and forcing a repair. The mechanical arm demonstrates that responsibility attribution is possible without LLM comprehension on 62 of 66 cases, with zero false positives on the 22 negative

Table 1: Baseline and ablation comparison: root success and unnecessary-reframe rates with 95% Wilson score intervals

System	Role	ALL root_success	HIDDEN_SHIFT root_success	NC unnecessary reframe
ORION full	SUBJECT	0.0208 [0.0037, 0.1090]	0.0312 [0.0055, 0.1574]	0.0000 [0.0000, 0.1936]
Live provider	BASELINE	0.0000 [0.0000, 0.0741]	0.0000 [0.0000, 0.1072]	0.0000 [0.0000, 0.1936]
<i>Baselines</i>				
Static ReAct workflow	BASELINE	0.0208 [0.0037, 0.1090]	0.0312 [0.0055, 0.1574]	0.0000 [0.0000, 0.1936]
Tree-search research	BASELINE	0.0208 [0.0037, 0.1090]	0.0312 [0.0055, 0.1574]	0.1875 [0.0659, 0.4301]
AREX-like	BASELINE	0.0208 [0.0037, 0.1090]	0.0312 [0.0055, 0.1574]	0.0000 [0.0000, 0.1936]
audit/follow-up				
SCION-like	BASELINE	0.0208 [0.0037, 0.1090]	0.0312 [0.0055, 0.1574]	0.0000 [0.0000, 0.1936]
dependency plan				
Iris-like information	BASELINE	0.0208 [0.0037, 0.1090]	0.0312 [0.0055, 0.1574]	0.0000 [0.0000, 0.1936]
state				
<i>Ablations</i>				
No explicit <i>W</i>	ABLATION	0.0208 [0.0037, 0.1090]	0.0312 [0.0055, 0.1574]	0.0000 [0.0000, 0.1936]
No explicit <i>M</i>	ABLATION	0.0208 [0.0037, 0.1090]	0.0312 [0.0055, 0.1574]	0.0000 [0.0000, 0.1936]
Generic retry	ABLATION	0.0208 [0.0037, 0.1090]	0.0312 [0.0055, 0.1574]	0.1875 [0.0659, 0.4301]
Full reset	ABLATION	0.0208 [0.0037, 0.1090]	0.0312 [0.0055, 0.1574]	0.0000 [0.0000, 0.1936]
No mechanic self-audit	ABLATION	0.0208 [0.0037, 0.1090]	0.0312 [0.0055, 0.1574]	0.1875 [0.0659, 0.4301]

Note: N=48 cases per system, 5 stochastic repeats, 2880 records total. 0 CANNOT_CHECK after live-provider recovery. The AREX-like arm is the highest-root-success baseline. All systems except the live-provider arm share one perception layer and are resource-matched. Paired percentile bootstrap (10k resamples) was used for comparisons; Holm correction was applied to secondary families. Full system IDs, per-family results, and mechanistic metrics are archived in `results/P1-T2_baseline_ablation_results.md`.

controls. Four general relations cover 26 cases; the remaining attributions rest on narrower rules whose generality is untested.

6.7.2 What the local evidence cannot establish

The historical N=48 falls short of the required 385 for H1 and 2,401 for H2. The achieved tier BELOW_TIER_D is below the minimum standard (TIER D), so this V1 study remains yes for its broad primary hypotheses. Within-study comparisons are populated (all systems below 0.03 root success; Table 1), but they do not authorize external superiority. The successor study is reported separately and is never pooled with these records.

6.7.3 Failure modes

The prospective failure taxonomy (Table 2) is populated from the frozen 2880-record archive. Abstention dominates across systems (81.4% of scored trials), with wrong-responsibility attributions at 9.2% and missed reframes at 7.5%; unnecessary reframes occur only under generic retry, tree search, and no-self-audit variants. The local suite additionally identified blind-responder shortcuts in template phrasing, filename-based family separation, and arithmetic defects in case resources, documented in the solvability audit artifact.

6.7.4 Limitations of the local suite

The local falsifier uses constructed cases with frozen semantic properties. It cannot establish open-world adequacy or external superiority. Template-level leaks (perfect separation on fixed phrases like “The team’s framing is”) were identified and documented but not repaired in this iteration.

Table 2: Failure taxonomy: roll-up by failure mode across all 12 systems and 2880 scored trials

Failure mode	Trials	Systems	Share	Representative blinded cases
ABSTENTION	2345	11	0.8142	P1-C-14aefdd887, P1-C-239f672ebc
WRONG_RESPONSIBILITY	265	4	0.0920	P1-C-239f672ebc, P1-C-43c471c226
MISSED_REFRAME	215	12	0.0747	P1-C-14aefdd887, P1-C-2a87cda138
UNNECESSARY_REFRAME	45	3	0.0156	P1-C-8b5889326e, P1-C-9fbf65695c
INCOMPLETE_REOPEN	10	1	0.0035	P1-C-557ed886fe, P1-C-ffb30e12b4

Note: One primary failure mode per trial by precedence (authority > invariant > trace integrity > abstention > control selectivity > diagnosis > targeting > reopening > terminal outcome).

Case IDs are blinded pseudonyms derived from the host-owned suite fingerprint; the raw suite is never published. Abstention dominates across all systems (81.4%). Wrong-responsibility attributions (9.2%) and missed reframes (7.5%) are the next most common; unnecessary reframes occur only under generic retry, tree search, and no-self-audit variants.

Half the mechanical arm attributions rest on singleton rules that may function as case-specific recognizers.

7 Powered mutation-necessity successor study

The 66-case study above remains a historical local falsifier and is not retrospectively promoted. A separately frozen successor, **P1.epistemic-mutation-necessity.v2.2.4**, tests the narrower residual left after nearest-work saturation: whether a high-level scientific mutation should require lower-level exclusion, a same-task counterfactual effect, preservation of a protected sibling or invariant, and an exact dependency-impact binding. Intervention-supported attribution, active diagnosis, diagnosis-to-valid-recovery admission, counterfactual inclusion-minimal repair, and selective dependency rollback are treated as donor-owned substrate rather than ORION novelty. In particular, REFLECT (arXiv:2606.09071), Causal Agent Replay (arXiv:2606.08275), R2Act (arXiv:2607.04623), DARC (arXiv:2608.11772), HERALD (arXiv:2608.06012), CausalFlow (arXiv:2605.25338), Causal-Repair (arXiv:2608.10613), and dependency-guided rollback (arXiv:2608.10502) determine the parent and ablation structure. The round-by-round dispositions and extracted structures are retained in the P1 donor-assimilation and engulfment ledgers.

7.1 Pre-bound design and integrity boundary

Each run contains 2,882 credential-free mechanical worlds: 480 hidden formulation/search-universe shifts and 2,402 evidence-only or execution-only negative controls. The hidden set is balanced across four shift families; the control size was fixed prospectively for the safety/non-regression estimand. Every arm receives the same public repair and diagnostic menu and the same four-unit intervention budget. Protected response matrices, family identity, gold minimal repairs, protected siblings, and dependency-impact sets remain host-only.

Nine runnable arms include static refusal, immediate outcome-flip and cost-greedy repair, REFLECT-like and CAR-like replay, active-VOI recovery, DARC/R2Act plus dependency rollback, a CausalFlow/HERALD-style minimal-counterfactual parent, and full ORION. Five direct ablations remove lower-level exclusion, protected-sibling checking, dependency-impact binding, K/W/M level ordering, or the intervention budget. The three strongest parents are all co-primary comparators. The registered support vector also requires zero unnecessary high-level reframes on controls,

whole-grid non-inferiority, zero protected-sibling regressions, zero forbidden high-level mutations, and mean cost no greater than the maximum strong-parent mean plus one unit.

The original execution receipt bound a replay envelope rather than the committed world receipt and therefore failed before any arm executed. A first additive amendment corrected that envelope. Review then found that the cost implementation compared ORION with every parent although the frozen rule names the maximum parent mean; no arm or terminal had been observed, so a second additive amendment bound the exact registered implementation without changing worlds, hypotheses, parents, seeds, margins, or protocol. All earlier receipts remain immutable.

7.2 Primary and disjoint replication results

The primary run and the prospectively bound disjoint replication both return `P1_MUTATION-NECESSITY_SUPPORTED`. Full ORION attains protected root-task success 1.0000 on the 480 hidden-shift worlds in each run. Each strong parent attains 0.4938 in the primary and 0.4833 in replication. Thus the paired advantage is 0.5063 in the primary and 0.5167 in replication; every 95% interval excludes zero and exceeds the registered 0.10 practical margin (Table 3). Exact paired p -values pass the three-comparison Holm family in both runs.

Table 3: Powered P1 v2.2.4 hidden-shift contrasts. Rates are protected root-task success over 480 pre-bound hidden-shift worlds per run. Intervals are paired percentile-bootstrap 95% intervals; all three paired tests pass the frozen Holm family in each run.

Run	Strong parent	ORION	Parent	Difference [95% CI]	Exact p
Primary	Active-VOI recovery	1.0000	0.4938	0.5063 [0.4625, 0.5521]	1.41×10^{-73}
Primary	DARC/R2Act + rollback	1.0000	0.4938	0.5063 [0.4604, 0.5500]	1.41×10^{-73}
Primary	CausalFlow minimal repair	1.0000	0.4938	0.5063 [0.4625, 0.5521]	1.41×10^{-73}
Replication	Active-VOI recovery	1.0000	0.4833	0.5167 [0.4708, 0.5604]	4.42×10^{-75}
Replication	DARC/R2Act + rollback	1.0000	0.4833	0.5167 [0.4729, 0.5604]	4.42×10^{-75}
Replication	CausalFlow minimal repair	1.0000	0.4833	0.5167 [0.4708, 0.5625]	4.42×10^{-75}

On 2,402 controls per run, ORION makes zero unnecessary formulation/search-universe re-frames and matches the active-VOI parent’s protected success at 1.0000; the paired difference and interval are exactly 0. Across all 2,882 worlds, ORION protected success is 1.0000. The primary whole-grid advantages over active-VOI, DARC/R2Act, and CausalFlow are 0.0843, 0.3220, and 0.9056; replication gives 0.0861, 0.3362, and 0.8980. ORION has zero protected-sibling regressions and zero forbidden high-level mutations in both runs. Its mean intervention cost is 1.8341 units in the primary and 1.8359 in replication, below the maximum strong-parent mean of 4.0.

The registered ablations move the mechanism, not only its label (Figure 2). Removing the protected-sibling check reduces hidden-shift protected success to 0.4938/0.4833 and introduces whole-grid sibling-regression rates 0.0843/0.0861. Removing dependency-impact binding reduces hidden-shift protected success to zero. Removing lower-level exclusion or K/W/M ordering creates forbidden high-level mutations on approximately 0.417 and 0.411 of primary worlds, respectively; the same failure directions recur in replication.

7.3 Independent verification and authorized claim

An independently implemented standard-library verifier imports neither the candidate policies, the production scorer/statistics, nor ORION licensing. It recomputes all 40,348 arm/world score rows per run, the complete analysis object, and the terminal. Both verification records report zero score

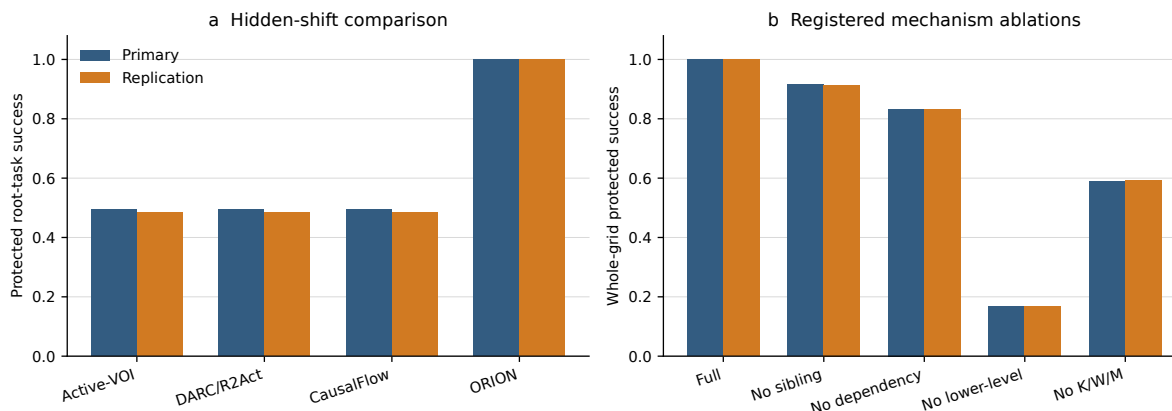


Figure 2: Descriptive rates from the pre-bound primary and disjoint replication. Left: protected success on 480 hidden-shift worlds. Right: whole-grid protected success for full ORION and registered mechanism ablations. Confirmatory inference uses the paired intervals and gates in Table 3, not visual bar separation.

mismatches, zero analysis mismatches, matching supported terminals, and PASS. Compressed raw results, public worlds, protected matrices, execution receipts, provenance, and SHA-256 manifests are archived under [research/revival/p1/confirmatory/v2.2/](#).

The result supports a bounded credential-free mechanical claim: on this frozen family, the science-specific conjunction of epistemic-level necessity, protected-invariant preservation, and dependency-impact binding improves protected recovery over the strongest assimilated parents while preserving the registered controls. It does not establish model-general performance, open-ended scientific superiority, literature-search adequacy, or novelty of attribution, minimal repair, action admission, dependency rollback, or certificate enforcement.

8 Nearest-work and claim boundary

The nearest-work programme deliberately removes broad automation claims from Paper I. AI Scientist-v2 contributes progressive agentic/tree exploration and experiment-management mechanisms; AI Co-Scientist contributes specialized generation, reflection, ranking, evolution, proximity, and resource-allocation roles; Kosmos contributes long-horizon coordination through a structured world model; Agent Laboratory contributes staged research artifacts, human checkpoints, and cost accounting; systems engineering contributes recursive decomposition, interfaces, traceability, and verification planning; rational metareasoning contributes the principle that additional computation should be justified by expected downstream value.

The 2026 literature narrows the claim further. AREX explicitly alternates research with constraint-wise audit and targeted follow-up [20], so recursive follow-up is not a standalone novelty. SCION compiles scientific intent into staged objectives, dependencies, verification checkpoints, artifacts and fallback conditions [32], so structured/dependency-aware scientific planning is not novel. Iris maintains an evolving information state containing revisable claims and uses epistemic actions to acquire decision-critical information [12], so Paper I cannot rely on “explicit evolving knowledge state” as a broad novelty statement. SciAgentArena provides realistic stepwise scientific-agent evaluation and demonstrates that open-ended autonomy remains difficult [18].

ORION may absorb or adapt those mechanisms. The principle of undiscovered public knowledge

[26] motivates the search-universe model W_t as a first-class state: relevant connections may exist across domain boundaries that no single query or prompt would retrieve. The capture-mark-recapture technique [15] and open-world evaluation for diverse perspectives [6] inform the coverage and stopping rules for W_t exploration. A systematic mechanism-matrix pass (Table 4) evaluates each nearest work against the four candidate deltas P1.D1–P1.D4 defined in the foundations section.

8.1 Per-delta verdicts

P1.D1 (K/W/M separation) — composition-only. Iris implements the K half completely: revisable claims carrying scope, status, and supporting/refuting evidence pointers, with Add/Update/Invalidate operators [12]. SCION’s Research Execution Plan plus layered epistemic memory covers explicit method state, and Kosmos covers a persistent structured world model [32, 21]. The only component without a verified parent is W —a separate model of the relevant search universe, distinct from knowledge about the task, that is itself revisable. Bisht et al. independently recommend building one as future work [3], confirming the gap is real but meaning ORION cannot claim to have identified it. A single unparented organ inside a three-part state is not an independent delta.

P1.D2 (typed formulation revision) — survives, narrowly. ARTS already diagnoses implementation-fault versus bad-hypothesis [14]. so the binary execution-versus-hypothesis split is struck. Iris already attributes a failure to which specific claim in the knowledge state is wrong [12]. MAST classifies multi-agent failures into 14 modes across three categories (system design, inter-agent misalignment, task verification); none of the 14 mode names targets a formulation, search-universe, representation, decomposition, or measurement-model coordinate [4]. Who&When and Who&When Pro annotate trajectories with the failure-responsible agent and decisive step [30, 19]. Auditable Agents names responsibility attribution as a surveyed dimension [23]. What remains unparented across all of these is that the *type* of the responsibility functions as an **authority gate**: a formulation or search-universe diagnosis selects which coordinate may be rewritten, and an evidence-only or execution-only diagnosis withholds the rewrite licence entirely. Every neighbour above produces a *descriptive label*; none attaches a *permission* to it. ORION has local mechanical evidence that this distinction bites—the hidden-shift falsifier exposed an over-broad reframe gate in which a singular responsibility diagnosis was wrongly treated as sufficient licence, and the repair is recorded in the claim ledger.

P1.D3 (scoped reopening) — struck. EviGraph performs dependency-directed reopening inside a 2026 autonomous research agent: it identifies a weak node, forms a repair group containing that node and its downstream subordinates, removes the subordinates, repairs the node, and regenerates affected nodes in dependency order [25]. EA-Graph adds content-hash anchoring, staleness detection independent of evidence strength, and claim withdrawal under upstream drift [13]. Beneath both, Doyle’s justification-based truth maintenance invalidates every dependent conclusion when a justification is retracted [11]; change impact analysis, regression test selection, and incremental build systems have done transitive-closure invalidation in engineering practice for decades. ORION therefore claims no independent novelty for scoped reopening. The residual is only the *trigger and scope*: EviGraph reopens from an evidence-graph inconsistency while holding Problem and Gap nodes immutable, whereas ORION reopens because a formulation or search-universe coordinate changed. That residual is inseparable from P1.D2 and belongs to the composition.

P1.D4 (mechanic self-audit) — composition-only. AREX already alternates research with recursive constraint-wise audit and emits targeted follow-up research from what the audit finds [20], which is the “audit generates further research” loop in full. The NASA Systems Engineering Handbook treats an empty cell in a requirements verification matrix as a defect that obliges new work, which is the “missing dimension mechanically generates an obligation” rule in full [22]. Auditable Agents establishes agent self-audit as a named, surveyed property [23]. What is left—that the audited object is ORION’s own research mechanics and that the emitted obligations are typed research questions rather than test procedures—is a domain transfer of two established disciplines, not a mechanism. It is presented as engineering substrate enabling the P1.D2 claim to be inspected and reproduced.

8.2 Residual claim

After the initial matrix, the mechanism itself was exposed to further nearest-work rounds before any confirmatory world was executed. REFLECT and Causal Agent Replay absorb intervention-supported attribution and causal replay; R2Act and DARC absorb diagnosis-conditioned recovery admission; HERALD and CausalFlow absorb counterfactual outcome-flip auditing and inclusion-minimal repair; dependency-guided rollback absorbs selective replay with preservation of independently supported state; and CausalRepair contributes dual static/dynamic causal-context slicing. Certificate/broker work separately absorbs proposal/admission/execution separation and scoped enforcement. These donors are not citation-only exclusions: their structures appear in the executable substrate, strongest-parent policies, and direct ablations. Rounds G and H found no material change to the post-engulfment design and are recorded as a bounded fixed point, not universal bibliographic completeness.

The smallest surviving and empirically tested Paper I claim is therefore:

In the frozen mechanical world family, after generic attribution, active diagnosis, admissible recovery, counterfactual minimal repair, dependency rollback, and certificate enforcement are granted to the parents, a scientific high-level K/W/M mutation is accepted only when lower-level alternatives have been excluded, the candidate causes a same-task recovery, a protected sibling or invariant is preserved, and the observed dependency impact matches the proposed reopen scope. This science-specific protected necessity composition improves recovery over every assimilated strong parent while withholding high-level mutation on evidence/execution controls.

Everything else has a parent. The state representation is Iris’s; the reopening machinery is Evi-Graph’s, EA-Graph’s, Doyle’s, dependency-guided rollback and change-impact analysis’s; recursive audit is AREX’s; the missing-cell obligation is the requirements verification matrix’s; attribution and replay are REFLECT/CAR territory; admission is R2Act/DARC territory; and minimal counterfactual repair is HERALD/CausalFlow territory. “Reformulation” itself has decades of history in automated planning [1]; metareasoning budgets are formalised elsewhere [8, 10]; and the gap is independently identified by Arbor [5] and Bisht et al. [3]. The bounded residual is a science-specific composition of protected necessity conditions, not ownership of any ingredient.

8.3 Open literature checks

Table 4: Nearest-work mechanism matrix: 36 mechanisms, dispositions, and their effect on ORION’s claim. Every arXiv identifier was resolved to a live abstract page and cross-checked for title/author/date before entry; rows marked [body-checked] were verified against the paper’s full text. Full source: `evidence/NEAREST_WORK_MATRIX_V2.md`.

Mechanism	Nearest work	What it does	Disp.	Effect on ORION’s claim
Recursive constraint-wise audit with targeted follow-up	AREX [20] (Lu et al., 2026)	Recursively refines answers via constraint-wise checks and targeted research iterations	ADOPT	Strikes “recursive audit + targeted follow-up” from novelty
Compact persistent research state	AREX [20]	Compresses long-horizon interaction history into retained state across research iterations	ADOPT	Strikes compact persistent research state from novelty
Staged dependency-aware Research Execution Plan	SCION [32] (Zheng et al., 2026)	Compiles intent into staged objectives, dependencies, verification checkpoints, fallbacks, layered epistemic memory	ADOPT	Strikes structured/dependency-aware planning; full-text check finds procedural rollback, not typed epistemic reopening
Target-conditioned hidden-target inverse search	SCION [32]	Inverse search conditioned on a target; batch active search under finite budgets	DEFER	Hidden <i>target</i> inside a fixed space $\neq$ wrong <i>search space</i> ; body unverified
Evolving information state of revisable claims [body-checked]	Iris [12] (Fu et al., 2026)	Claims with scope, evidence-pointer sets, status; Add/Update/Invalidate operators	ADOPT	Strikes the <i>K</i> half of P1.D1 outright
Epistemic actions decoupled from solution modification [body-checked]	Iris [12]	Discriminative evidence acquisition without modifying the retained solution	ADOPT	Confirms evidence responsibility routes to acquisition, not rewrite—the repair ORION’s own falsifier forced
Attribution of failure to the specific wrong claim [body-checked]	Iris [12]	Localizes which belief is wrong and rewrites it	ADOPT	Narrows P1.D2: only typing by formulation coordinate is left
Faulty-implementation vs. bad-hypothesis diagnosis	ARTS [14] (Juneja et al., 2026)	Inspects prior logs, diagnoses implementation-vs-hypothesis faults, navigates (not restructures) hypothesis space	ADOPT	Strikes the binary responsibility split from P1.D2
Downstream dependency tracing and regeneration [body-checked]	EviGraph [25] (Ren et al., 2026)	Repair group = weak node + downstream subordinates; regenerates dependents in order; Problem/Gap nodes immutable	ADOPT	Strikes P1.D3 as an independent delta; excludes reopening from problem-statement change
Content-anchored staleness with claim withdrawal [body-checked]	EA-Graph [13] (Hsu et al., 2026)	Content-hash anchoring; unaffected / affected / <i>unprovable</i> outcomes	ADOPT	Reinforces the P1.D3 strike; prior art for ORION’s fail-closed semantics
Dependency-directed backtracking; justification-based TMS	Doyle, AIJ 1979 [11]	Invalidates every dependent of a retracted justification	ADOPT	Confirms scoped invalidation is a named 1979 mechanism
Assumption-based TMS over environments	de Kleer, AIJ 1986 [7]	Multi-context coexistence via assumption sets	COMPOSE	Multi-context form for provisional reframes
Minimal-change belief revision, entrenchment	AGM, JSL 1985 [2]	Contraction/revision ordered by epistemic entrenchment	ADAPT	Formal parent of ORION’s <i>K</i> -revision semantics

Table 4 (continued): nearest-work mechanism matrix

Mechanism	Nearest work	What it does	Disp.	Effect on ORION’s claim
Change impact analysis over trace links	Systems-engineering CIA	Traces impact through requirements traceability links	ADOPT	Engineering form of scoped invalidation
Regression test selection / incremental-build invalidation	STARTS, ASE 2017	Impacted tests via transitive closure of changed types	ADOPT	Second parent-discipline strike on P1.D3
Verification matrix: empty cell is a defect	NASA SE Handbook [22]	Bidirectional traceability; no empty cells; missing relation obliges new verification	ADAPT	Strikes the generic P1.D4 rule; variant audits own mechanics, emits research questions
Recursive decomposition with interfaces and V&V	NASA SE Handbook [22]	Recursive decomposition; interface fixing; parent/child V&V transport	ADOPT	Recursive engineering decomposition is not a delta
Named auditability dimensions incl. responsibility attribution	Auditable Agents [23] (Nian et al., 2026)	Five auditability dimensions incl. responsibility attribution	COMPOSE	“The agent audits itself” is not ORION’s contribution
Attribution to responsible agent and decisive step	Who&When [30]; Who&When Pro [19]	Annotates trajectories; attribution target is agent identity + step	COMPOSE	Attribution per se excluded; formulation-coordinate typing stays open
Empirical 14-mode failure taxonomy	MAST [4] (Cemri et al., 2025)	14 modes over system design, inter-agent misalignment, task verification	COMPOSE	Full-text mode check finds descriptive labels but no formulation-coordinate rewrite permission
Progressive agentic tree search + experiment manager	AI Scientist-v2 (Yamada et al., 2025)	End-to-end hypothesis→experiment→manuscript cycle	ADOPT	Tree search and automated science excluded from novelty
Role-specialized generate/debate/evolve + tournament ranking	AI Co-Scientist [21] ^a (Gottweis et al., 2025)	Generation, reflection, ranking, evolution, proximity, meta-review roles	ADAPT	Multi-role reasoning not a delta; tournament standing is never scientific authority
Structured world model shared across agents	Kosmos [21] (Mitchener et al., 2025)	Shares a structured world model across analysis and search agents	ADOPT	Long-horizon coordination excluded; objective revisability unestablished
Persistent hypothesis tree, falsified-subtree pruning [body-checked]	Arbor [5] (Chen et al., 2026)	Coordinator + short-lived executors; prunes falsified subtrees; no cascading invalidation	COMPOSE	External confirmation the formulation gap is real; required H1 baseline
Position: AI scientists as co-scientists only	Bisht et al. [3] (2026)	Recommends world models representing shifting objectives	COMPOSE	Independently names ORION’s target gap; recommends the <i>W</i> model
Reformulation as change of problem representation	Alarnaouti et al. [1] (2023)	Solution-preserving re-representation for easier planning search	ADAPT	The word “reformulation” carries no novelty; ORION’s is wrong-formulation-triggered
Rational metareasoning / value of computation	De Sabbata et al. [8] (2024)	Value-of-Computation reward; Expert Iteration	ADOPT	Cost-aware stopping excluded from novelty
Bounded deliberation with value-of-information halting	Cognitive Friction [10] (Di Gioia, 2026)	Rollout VoI; net-utility halting condition for tool-using agents	ADOPT	Agent-setting VoI control absorbed

Table 4 (continued): nearest-work mechanism matrix

Mechanism	Nearest work	What it does	Disp.	Effect on ORION’s claim
Realistic stepwise-verified scientific-agent evaluation	SciAgentArena [18] (Liu et al., 2026)	200 research tasks; stepwise verification; novelty deficit in open-ended questions	ADOPT	External task source; empirical premise for the hidden-shift hypothesis
Staged research pipeline, checkpoints, cost accounting	Agent Laboratory (Schmidgall et al., 2025)	Review → experiment → report with human feedback and cost measurement	ADOPT	Stage receipts and cost accounting absorbed as protocol requirements
Research-question certificate	FirstResearch (Wang, 2026)	Certificate making a research question inspectable	REJECT	Withdrawn by the author 28 Jul 2026; route closed
Post-hoc dependency and execution graph	GRADE [27] ^b (Zhao, 2026)	Run graph with execution and dependency edges for diagnosis	DEFER	No runtime invalidation or reopening; no delta effect
Critic agent: causal questioning and falsification	AHOIS [16] ^c (Zeng et al., 2026)	Interrogates hypotheses via causal questioning, counterexamples, falsification criteria	COMPOSE	Falsification-criteria generation is distinct from typed coordinate revision
Project-agent-resource collaboration layer	Clarus (Guo et al., 2026)	Multi-phase auditable research-process organization	DEFER	Organizational layer; no delta effect
Aligned state checkpoints with rewind/resume	AgentRewind [33] (Zhuang et al., 2026)	Snapshots agent context + controlled environment; rewinds to recoverable state	DEFER	State recovery to a checkpoint, no typed failure-responsibility taxonomy; no P1.D2 effect
Executable-state transition with re-anchoring	ScienceFlow [31] (Zhao et al., 2026)	Re-anchors ML-research agent’s executable state under evidence-aware controller	DEFER	Rebuilds <i>executable state</i> , not a formulation-coordinate rewrite gated by typed responsibility; no P1.D2 effect

Dispositions: **ADOPT** 19 (mechanism absorbed; stripped from novelty) · ADAPT 4 (mechanism absorbed with a stated variant) · COMPOSE 7 (constrains the composition, not a parent) · DEFER 5 (checked, not close; body unverified) · REJECT 1 (withdrawn). Row total: 36.

Verification gaps (stated, not papered over): (1) SCION checkpoint-failure reopening semantics—closed by full-text reading: runtime rollback is procedural-anomaly gated, not epistemic-formulation-change gated; EviGraph independently strikes P1.D3, so the gap changes no verdict; (2) MAST mode names—retrieved from full text (Appendix A); none targets a formulation coordinate; P1.D2 narrowing does not tighten further; (3) surfaced-but-unfetched cluster—retrieved and read; none attaches a permission to modify the formulation or search universe; P1.D2 safe against that family; (4) fresh literature-closure pass—completed 2026-08-17: 8 papers from the 2608 cs.AI series checked; none attaches a responsibility-typed reformulation/authority licence; P1.D2 unaffected; (5) absence claims scoped to the stated function-only search space, not exhaustive 2026 coverage.

^a The AI Co-Scientist identifier (arXiv:2502.18864, *Towards an AI co-scientist*, Gottweis et al., 2025) was verified; a journal version exists unverified and is not cited. ^b GRADE [27] is cited in the bibliography as span-level error localization (arXiv:2606.02060); the GRADE arXiv ID 2606.22741 is recorded in the matrix source. ^c AHOIS (arXiv:2606.26722) is cited in the bibliography as self-improving error diagnosis (arXiv:2604.17658); both IDs recorded in the matrix source.

The two pre-submission checks named in ‘JOURNAL_READINESS.md’ are complete. MAST’s 14 mode names were retrieved from the full text (Appendix A); none targets a formulation, search-universe, representation, decomposition, or measurement-model coordinate, so the P1.D2 narrowing

does not tighten further [4]. The failure-attribution cluster that previously bounded the P1.D2 verdict was retrieved and read in full: TRAIL contributes a formal error taxonomy for agentic workflow traces but stops at evaluation [9]; REFLECT attributes silent failures to candidate error steps and validates them by controlled replay [17]; and the span-level/DRIFT and self-improving-diagnosis/ErrorProbe frameworks localize error spans and responsible agents while leaving the formulation itself untouched [27, 16]. None of the cluster attaches a permission to modify the formulation or search universe on the basis of the diagnosed error type, so the P1.D2 verdict is no longer provisional on that route.

The fresh literature-closure pass named in ‘JOURNAL_READINESS.md’ is complete. Eight papers from the 2608 cs.AI series were surfaced and checked against the P1.D2 claim: AgentRewind provides aligned state checkpoints with rewind/resume for long-horizon agents but has no typed failure-responsibility taxonomy [33]; ScienceFlow re-anchors a long-horizon ML-research agent’s executable state under an evidence-aware controller but rebuilds *executable state*, not a formulation-coordinate rewrite gated by typed responsibility [31]; Demystifying Agent Skills, Drift Diagnosis, Wrong but Useful, Session Handover, SAFARI, and Past and Future of AI Scientists each address related topics (agent skill characterisation, drift diagnosis, skill utility, session transfer, and AI-scientist survey respectively) but none attaches a responsibility-typed reformulation or authority licence. AgentRewind and ScienceFlow are the closest surfaced works—both involve state recovery or re-anchoring—and are recorded as DEFER rows in the mechanism matrix (Table 4). The P1.D2 verdict is unaffected by this pass. The SCION checkpoint-failure semantics question raised in an earlier pass has been closed by reading the full paper text: SCION distinguishes precompiled fallback conditions in the Research Execution Plan from a runtime rollback mechanism in which critic checkpoints trigger rollback, branch termination, re-routing, or resource reallocation when an agent drifts, hallucinates, or enters an unproductive branch [32]. The runtime rollback is therefore gated on procedural anomalies rather than on the type of responsibility for a failure, and SCION does not reopen already-completed dependent stages conditional on a responsibility type; that residual trigger remains ORION-specific and is unaffected.

9 Limitations and open empirical boundary

The present repository establishes an implemented reference architecture, local deterministic falsifiers, and a powered credential-free mechanical comparison with disjoint replication. It does not establish reliable open-web retrieval, general scientific-language interpretation, open-world parent-discipline recall, model-general superiority, or autonomous method improvement.

Recent autonomous-research benchmarks report substantial remaining difficulty in wide literature discovery and end-to-end scientific rediscovery [28, 29]. We treat those results as motivation for hostile external evaluation, not as evidence that ORION solves the problem.

Several boundaries are especially important.

Local worlds are not external science. The successor study uses fresh pre-bound seeds, protected matrices, three strong assimilated parents, and independent rescoring, but its worlds are generated mechanical response environments. Passing this family is evidence for the tested causal/authority composition, not evidence that an LLM research agent will recognize the same latent structure in open-ended scientific practice. Cross-domain, model-bearing evaluation remains future work.

Reconstruction quality is only partially measured. The current local suite checks key transition behaviors, but the deeper hypotheses about late-depth trace fidelity, reviewer localization time, and semantic mismatch require a dedicated prospective protocol.

Search and semantic adequacy are delegated, not solved here. Paper II owns route coverage/recall and Paper III owns scientific semantic projection and global reconstruction. A Paper-I success cannot be used as evidence that those capabilities work on real literature.

Verification remains separate. Search, proposal, diagnosis, and reframe mechanisms do not mint scientific authority. Paper IV owns the stronger evidence-binding and protected-verification claim.

Method change remains gated. Paper I can expose a possible method failure, but it does not authorize M_t to rewrite itself. Paper V owns failure attribution, challenger construction, replay/fresh transfer, protected assurance, and host promotion.

Literature closure is bounded rather than universal. Rounds A–H reached a fixed point for the exact post-engulfment residual and directly shaped the strongest parents. Because the 2026 repair-agent literature is moving rapidly, a final pre-submission delta search and replacement of preprints by archival versions remain required. No broad novelty statement follows from the bounded fixed point.

Observed limitations

The historical V1 arm remains underpowered. With $N=48$ cases, the original study achieves BELOW_TIER_D (below TIER D) and is yes for its original broad primary hypothesis. Those records, including the low root-success rates in Tables P1-2 and P1-3, are retained as negative history and are not pooled with the powered successor. The v2.2.4 result addresses a narrower prospectively frozen mechanical necessity claim on 2,882 worlds per run; it does not retroactively validate V1’s broader formulation.

Perfect success is generator-bounded. Full ORION attains protected success 1.0000 in both v2.2.4 runs. This is credible only at the level of the deterministic registered response generator: the candidate policy and gold structure share a deliberately explicit action/response vocabulary. The strong parents and hostile ablations show that target flipping, minimal repair, and dependency roll-back alone are insufficient under that vocabulary, but the ceiling result may not survive perceptual ambiguity, noisy interventions, or learned proposal generation.

Template-level leaks exist in the local suite. The solvability audit identified perfect separation on fixed phrases (“The team’s framing is” distinguishes all 44 hidden-shift cases from all 22 controls) and filename-based family separation (proposal/trial stems identify DECOMPOSITION cases). These leaks were documented but not repaired in this iteration.

Mechanical arm generality is untested. Half the correct attributions (32 of 66) rest on rules that fire on exactly one case. Only four general relations cover three or more cases. A fresh set of cases is required to distinguish general mechanisms from case-specific recognizers.

Blind-responder shortcuts affect half the suite. Thirty-three of 66 cases can be solved by a classifier that reads only resource path stems and two fixed template phrases. The remaining 33 PARENT/REPRESENTATION/MEASUREMENT cases require genuine arithmetic reasoning.

10 Conclusion

Paper I reframes recursive research as an epistemic-state problem rather than a synonym for repeated model calls. ORION separates the knowledge state, relevance/search-universe model, and governed method state; diagnoses typed residuals so that responsibility determines authority to rewrite a formulation coordinate; and reopens only closure that depends on a changed coordinate. Recursive audit, active diagnosis, counterfactual minimal repair, diagnosis-to-action admission, certificate enforcement, and dependency rollback are absorbed donor substrate rather than independent deltas. The surviving tested composition requires lower-level exclusion, a high-level counterfactual effect, protected-invariant preservation, and exact dependency-impact binding before scientific K/W/M mutation.

The powered v2.2.4 successor supports that bounded composition on its credential-free mechanical family. Across a prospectively frozen primary run and disjoint replication, ORION exceeds all three strong parents by more than 0.50 protected hidden-shift success, preserves perfect control restraint and protected success, and passes every frozen non-regression, safety, multiplicity, and cost gate. Direct ablations reproduce the intended mechanism failures, and an independent implementation recomputes every score and analysis field without disagreement. These results justify a peer-review claim about the registered mechanical family. They do not justify model-general, open-ended, retrieval, or autonomous-science superiority, which remain explicit external boundaries.

11 Data and code availability

11.1 Repository access

The source code, protocol definitions, case suite, and repository-held evidence artifacts used by this revision are version-controlled at `SzeChunYiu/ORION`. Persistent DOI assignment remains pending publication. Result-bearing artifacts are identified by their repository paths and content bindings rather than by an unverified archival claim.

11.2 Reproducing the results

11.2.1 Powered v2.2.4 successor

The primary and replication directories under `research/revival/p1/confirmatory/v2.2/` each contain compressed public worlds, protected response matrices, raw results, result JSON, an independent verification record, provenance, and `SHA256SUMS`. The primary execution identity is `PRIMARY_EXECUTION_FREEZE_V3.json`; earlier receipts and both additive amendments are retained. The replication directory binds its own disjoint world and execution receipts. `PRIMARY_REPLICATION_CONCORDANCE.json` checks the predeclared cross-run conditions without pooling the runs.

The exact decompression, independent-rescoring, byte-comparison, and figure-regeneration commands are given in `REPRODUCE.md`. The independent verifier uses only the standard library and imports neither the candidate policies, production scorer/statistics, nor ORION licensing. It

recomputes 40,348 arm/world rows per run and must reproduce the complete analysis object and terminal.

11.2.2 Archived-result regeneration

```
make paper01-results
```

regenerates the Paper I publication tables from archived raw records only. It does not execute systems under test or create missing external evidence. On a checkout without an admissible archived study run, the underlying module returns `CANNOT_CHECK`; malformed or unbound archives are refused rather than converted into publishable numbers. The exact command, inputs, outputs, and exit semantics are documented in the paper’s repository-level `REPRODUCE.md`.

11.2.3 Clean environment

The repository is Python-based with `pyproject.toml` dependency specification. A fresh code/test environment requires:

- Python 3.11 or later;
- `pip install -e .[dev]` from the repository root; and
- model-provider access only for commands that actually execute a live provider arm.

The paper-specific reproduction note above, rather than a repository-root container recipe, is the current executable reproduction contract.

11.3 Runtime and cost

The credential-free v2.2.4 campaign and its independent verification each complete in approximately one minute on the current CPU environment. They make no network call and incur no model-provider cost. Algorithmic effort is fixed at 2,882 worlds, nine runnable arms, five ablations, and 10,000 registered paired-bootstrap resamples. Historical provider-oriented V1 cost remains `CANNOT_CHECK`; it is not imputed from the mechanical successor.

11.4 Licensing status

This revision does not contain a repository-level `LICENSE` file or a project license declaration in `pyproject.toml`. Redistribution terms for code, case definitions, and protocol artifacts are therefore `CANNOT_CHECK` from the repository state and must be resolved explicitly before archival/publication. Model-provider terms remain separately governed by the relevant providers.

11.5 Artifact fingerprinting

Result-bearing execution identities are bound through the protocol and archived records. Relevant coordinates include:

- dataset/suite revisions and fingerprints;
- baseline configuration hashes;
- evaluator identity;

- frozen split hashes; and
- the exact subject revision and provider/model revisions for a live run.

Reproduction of an empirical result requires the exact bound subject and admissible archived records; prose or a protocol declaration alone is not evidence that the run occurred.

12 Ethics, safety, and resources

12.1 Ethical considerations

This work studies mechanistic aspects of research workflow design rather than reporting a deployment in a human-facing domain. The frozen local evaluation uses constructed cases and does not require human-subject records or personally identifiable information. Any later external-provider or retrieval execution must be documented separately from this local evidence rather than inferred from the protocol design.

12.2 Safety considerations

ORION’s design explicitly separates authority promotion (Paper IV) from formulation reconstruction (Paper I). Paper I mechanisms can expose possible method failures but cannot authorize M_t to rewrite itself. The reframe gate is mechanically bounded: only formulation and search-universe coordinates may trigger a reframe, and evidence/execution-only negative controls exercise the corresponding non-reframe behavior in the frozen local suite.

12.3 Resource usage and environmental impact

The repository does not yet contain an admissible clean-environment resource ledger for the full Paper I campaign. Numerical model-token, runtime, monetary-cost, or carbon-impact claims are therefore `CANNOT_CHECK` in this revision. A future result-bearing run must retain actual resource usage under its frozen provider/environment bindings before those quantities are reported.

12.4 Dual-use considerations

The epistemic-state machinery described here is general-purpose and could be applied outside scientific research. This paper evaluates scientific-workflow mechanics and does not present an empirical evaluation of military or other high-risk deployment applications. The authority boundaries above are part of the implemented design, but they should not be read as evidence of safety in domains that were not evaluated.

References

- [1] Lama Alarnaouti, George Baryannis, and Mauro Vallati. Reformulation in automated planning: A survey. *arXiv preprint arXiv:2301.10079*, 2023.
- [2] Carlos E. Alchourrón, Peter Gärdenfors, and David Makinson. On the logic of theory change: Partial meet contraction and revision functions. *The Journal of Symbolic Logic*, 50(2):510–530, 1985.

- [3] Aditya Bisht, Raj Kumar, Kevin Maik Jablonka, Mausam, and Sriram Krishnan. Are ai scientists actually scientists? a critical assessment of autonomous discovery systems. *arXiv preprint arXiv:2605.08956*, 2026.
- [4] Mert Cemri et al. Why do multi-agent llm systems fail? *arXiv preprint arXiv:2503.13657*, 2025.
- [5] Hao Chen et al. Arbor: Persistent hypothesis trees for autonomous scientific research. *arXiv preprint arXiv:2606.11926*, 2026.
- [6] Hung-Ting Chen and Eunsol Choi. Open-world evaluation for retrieving diverse perspectives. In *Proceedings of NAACL 2025*, 2025.
- [7] Johan de Kleer. An assumption-based tms. *Artificial Intelligence*, 28(2):127–162, 1986.
- [8] Stefano De Sabbata, Theodore Sumers, Badr AlKhamissi, Antoine Bosselut, and Thomas L. Griffiths. Rational metareasoning: Value of computation for llm reasoning budgets. *arXiv preprint arXiv:2410.05563*, 2024.
- [9] Darshan Deshpande, Varun Gangal, Hersh Mehta, Jitin Krishnan, Anand Kannappan, and Rebecca Qian. TRAIL: Trace reasoning and agentic issue localization. *arXiv preprint arXiv:2505.08638*, 2025.
- [10] Marco Di Gioia. Cognitive friction: Bounded deliberation with value-of-information halting for tool-using agents. *arXiv preprint arXiv:2603.30031*, 2026.
- [11] Jon Doyle. A truth maintenance system. *Artificial Intelligence*, 12(3):231–272, 1979.
- [12] Shaokang Fu et al. Beyond solution-centric search: Adaptive inquiry and knowledge revision for autonomous ml engineering. *arXiv preprint arXiv:2608.02143*, 2026.
- [13] Chih-Hsuan Hsu, Alexander Chi, and John Everett. Ea-graph: Content-anchored staleness detection for verification claims in autonomous research. *arXiv preprint arXiv:2608.04278*, 2026.
- [14] Siddhartha Juneja, Abhishek Jain, Deepak Nathani, He Wang, and Xuan Wang. Arts: Automated research trajectory search for hypothesis refinement. *arXiv preprint arXiv:2606.21891*, 2026.
- [15] Monika Kastner, Sharon E. Straus, K. Ann McKibbin, and Charlie H. Goldsmith. The capture-mark-recapture technique can be used as a stopping rule when searching in systematic reviews. *Journal of Clinical Epidemiology*, 62(2):149–157, 2009.
- [16] Jiazheng Li, Emine Yilmaz, Bei Chen, and Dieu-Thu Le. Towards self-improving error diagnosis in multi-agent systems. *arXiv preprint arXiv:2604.17658*, 2026.
- [17] Xiaofeng Lin, Yingxu Wang, Tung Sum Thomas Kwok, Daniel Guo, Sahil Arun Nale, Charles Fleming, and Guang Cheng. REFLECT: Intervention-supported error attribution for silent failures in LLM agent traces. *arXiv preprint arXiv:2606.09071*, 2026.
- [18] Tianyu Liu et al. Benchmarking ai agents for addressing scientific challenges across scales. *arXiv preprint arXiv:2606.12736*, 2026.

- [19] Ziyi Liu et al. Who&when pro: Advancing failure attribution in multi-agent systems. *arXiv preprint arXiv:2607.09996*, 2026.
- [20] Shuqi Lu et al. Arex: Towards a recursively self-improving agent for deep research. *arXiv preprint arXiv:2607.21461*, 2026.
- [21] Christopher Mitchener et al. Kosmos: Structured world models for scientific research agents. *arXiv preprint arXiv:2511.02824*, 2025.
- [22] NASA. *NASA Systems Engineering Handbook*. National Aeronautics and Space Administration, rev. 2 edition, 2016.
- [23] Tianyu Nian et al. Auditable agents: A framework for transparent and accountable ai systems. *arXiv preprint arXiv:2604.05485*, 2026.
- [24] David L. Parnas. On the criteria to be used in decomposing systems into modules. *Communications of the ACM*, 15(12):1053–1058, 1972.
- [25] Xuanyu Ren et al. Evigraph: Dependencies make ai scientist better. *arXiv preprint arXiv:2608.04738*, 2026.
- [26] Don R. Swanson. Undiscovered public knowledge. *The Library Quarterly*, 56(2):103–118, 1986.
- [27] Jiaming Wang, Ziteng Feng, Jiangtao Wu, Ruihao Li, Qianqian Xie, Yuxiang Ren, He Zhu, Xueming Han, Fanyu Meng, Junlan Feng, and Jiaheng Liu. Where do deep-research agents go wrong? span-level error localization in agent trajectories. *arXiv preprint arXiv:2606.02060*, 2026.
- [28] Lei Xiong et al. Autoresearchbench: Benchmarking ai agents on complex scientific literature discovery. *arXiv preprint arXiv:2604.25256*, 2026.
- [29] Wanghan Xu et al. Researchclawbench: A benchmark for end-to-end autonomous scientific research. *arXiv preprint arXiv:2606.07591*, 2026.
- [30] Yunqi Zhang et al. Which agent causes task failures and when? *arXiv preprint arXiv:2505.00212*, 2025.
- [31] Zhenguo Zhao et al. Scienceflow: A long-horizon agent for ml research, scientific discovery and beyond. *arXiv preprint arXiv:2608.14354*, 2026.
- [32] Yining Zheng, Yuxin Wang, Jiahao Lu, Shicheng Fang, Weiyi Wang, Yongzhuo Yang, Bowen Li, Haochen Ma, Chen Hu, Bowen Chen, Yang Wang, Huanhui Chen, Yitong Chen, Jiajun Chen, Zhiyuan Li, Yanlin Li, Zhuo Yang, Qifeng Wu, Jiaying He, Zhijie Jinluo, Xiaohu Xu, Yi Feng, Juncheng Qian, Yizhou Chen, Yang Cheng, Tong Zhu, Tianlei Ying, Hongyu Yu, Hongjun Xiang, and Xipeng Qiu. Rethinking scientific discovery in the agentic era. *arXiv preprint arXiv:2607.03863*, 2026.
- [33] Wenyu Zhuang, Xiaodong Chen, Shuyan Duan, Haiyi Zheng, Yiming Li, and Yating Zhang. Agentrewind: Recoverable execution for long-horizon llm agents. *arXiv preprint arXiv:2608.14380*, 2026.